

# Hydris AI — Basin Intelligence & Water Risk

DataCraft'26 · Problem Statement 1 · Project Report

## PROBLEM STATEMENT

Companies with factories worldwide cannot easily answer: “How water-risky is each of my sites, why, and how will that change?” Our goal: take a factory location, automatically resolve its watershed, and return a multi-risk profile (scarcity, flood, drought, groundwater) for today and 2030/2050 scenarios, with an explanation of what drives every score — aligned with the Positive Water Impact (PWI) framework of Availability, Quality and Accessibility.

## DATASET DESCRIPTION

- **Core source:** WRI Aqueduct 4.0 (CC-BY 4.0) via Google Earth Engine — baseline\_annual (13 indicators, e.g. baseline water stress, drought, riverine/coastal flood, groundwater decline) and future\_annual (optimistic/BAU/pessimistic × 2030/50/80), at HydroBASINS level-6 basin resolution.
- **Driver datasets (all probed live in GEE before use):** CHIRPS daily rainfall (30 yr), MODIS MOD16 evapotranspiration & MOD13Q1 NDVI, NASA GLDAS soil moisture/runoff, GRACE groundwater storage, SMAP soil saturation, HydroSHEDS flow/elevation, HYSOGs250m soil groups, ESA WorldCover, SoilGrids, JRC GHSL built-up; plus Nominatim geocoding and NewsAPI clippings.
- **Pilot set:** 5 factories on 4 continents (Chennai, Tiruppur, Fresno, Hamburg, Riyadh) × 3 driver engines = 15 driver profiles; any point on Earth can be added live.
- **Preprocessing:** Aqueduct −9999 → null (excluded, weights renormalized, never set to 0); +9999 kept as arid-mask; annual series built with a batched, gap-safe yearly() helper (handles MODIS pre-2000 and GRACE 2017–18 gaps); unit scaling; severities normalized to 0–1; GEE polygons repaired (GeometryCollection → Polygon) and simplified for the map.

## METHODOLOGY

- **1. Basin resolution & headline scores:** point-in-polygon lookup in Aqueduct returns 13 indicators, mapped to three PWI dimensions (Availability = scarcity composite  $0.36 \cdot bws + 0.36 \cdot bwd + 0.18 \cdot iav + 0.10 \cdot sev$ , flood, drought, groundwater; Quality =  $ucw, cep, rri$ ; Accessibility =  $mean(udw, usa)$ ). Future scenarios use WRI projections.
- **2. Driver engines (the “why”):** **Drought**  $D = 0.35 \cdot \text{Meteorological (gamma-fit SPI, SPEI-lite)} + 0.30 \cdot \text{Agricultural (soil moisture, NDVI)} + 0.35 \cdot \text{Hydrological (runoff, GRACE)}$ . **Groundwater**  $G = 0.35 \cdot \text{extraction pressure} + 0.30 \cdot \text{aquifer decline (GRACE trend)} + 0.20 \cdot \text{recharge weakness}$ . **Flood** =  $rain \cdot .25 + SCS\text{-}CN \text{ runoff} \cdot .25 + catchment \cdot .20 + soil \text{ saturation} \cdot .15 + urbanization \cdot .15$  (Hazard × Exposure × Vulnerability). Output: hazard 0–5 plus attribution % per driver.
- **3. Validated AI layer:** Groq Llama 3.3 70B (temperature 0) writes plain-language narratives and tags news sentiment; a numeral validator checks every number against the computed JSON, retries, then falls back to a deterministic template.
- **4. Product:** FastAPI backend + Supabase cache + React/MapLibre dark map with PWI ring gauges, 13 indicator bars, 2030/2050 toggle, custom weights and add-a-factory.

## RESULTS & INSIGHTS

- **Distinct risk fingerprints:** the driver engines separate sites that WRI’s single overall score lumps together. Chennai is flood-led (3.57/5, rainfall-driven), Fresno groundwater-led (3.97/5, extraction + GRACE decline), Riyadh groundwater-led (3.26/5).
- **Surfaced hidden risk:** Hamburg is “Low” overall (0.75/5) yet its drought hazard is 2.98/5 — a signal the overall score hides.
- **Explainability:** attribution sums to 100% for every site × risk (15/15); top drivers vary by site (SPEI, SCS-CN runoff, GRACE storage, extraction pressure).
- **Physical plausibility:** Chennai curve number ≈ 85–92 (urban); Fresno GRACE decline flagged high — consistent with known hydrology (validation gates 0, 1a–1c passed).
- **Honesty metrics:** 0 leaked −9999 values; missing data shown as “n/d” (Chennai groundwater, Riyadh drought), never 0; LLM text rejected if any number is absent from the computed JSON.
- **Performance:** cached profile ~100 ms; live driver run 2–47 s per risk on pilots (1–2 min for a brand-new site).

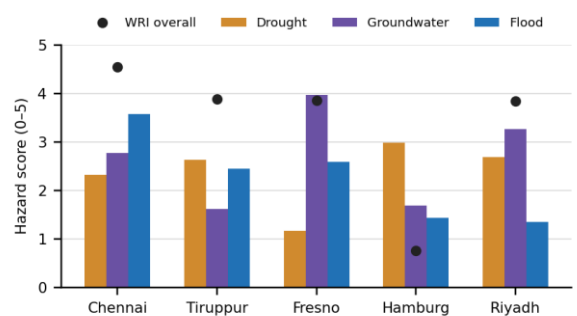


Fig. 1 — Driver-engine hazard vs. WRI overall score, 5 pilot sites

## NOVELTY

- **Scores that explain themselves:** each WRI score is paired with a satellite-computed driver breakdown, turning a number into an action (e.g. which share of flood risk comes from rainfall vs. impervious runoff).
- **Honesty layer by design:** every driver carries a source ID and confidence tag (computed / proxy / regional / no-data); GRACE always labeled regional; flood/drought/groundwater badged “not projected by WRI” instead of faked futures.
- **Hallucination-proof AI:** mechanical numeral validation of LLM output — AI explains, it never invents data.
- **Global, zero-storage, resilient:** computes on demand in Earth Engine for any coordinate; Supabase → bundled demo → live-compute fallback keeps the product working even if GEE is down.